

Lesson 4-1 · Period 1

Inverse Variation — Introduction

Klimesara-adapted · Student-centered · No Blooket

DOK 1

5 min

notice & wonder

Two rectangles — both have area = 144 sq units Rect 1: length = 72, width = 2 Rect 2: length = 24, width = 6

1. **Notice:** Write something you notice.
2. **Wonder:** Write something you wonder about rectangles with area 144.
3. **Sketch:** Draw a third rectangle with area 144. Label length and width.

DOK 1-2

12 min

teacher models

Inverse Variation Rules (keep on your page)

1. INVERSELY: their **product** is a constant k $xy = k$ or $y = k/x$
2. As one variable INCREASES, the other DECREASES proportionally.
3. To find k : multiply any given pair.
4. To find unknown: write $y = k/x$, then substitute.

TRAP: $y/x = k$ = DIRECT $xy = k$ = INVERSE

Launch — Example 1: Table Check

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DOK 1-2

12 min

teacher models

Is this an inverse variation?

x	1	2	3	5	6	15
y	25.5	12.75	8.50	5.10	4.25	1.70

Step: Compute xy for each column.

If every product equals the same number → **YES, inverse variation.**

Bridge: What was xy for your Do Now rectangles?

Launch — Example 2: Write the Equation

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DOK 1-2

12 min

teacher models

Given a pair of values, write the equation Two variables vary inversely. $x = 10$ when $y = 3$.

1. Find k : $k = xy = \underline{\hspace{2cm}}$
2. Write the equation: $y = k/x \rightarrow y = \underline{\hspace{2cm}}$
3. Find y when $x = -6$: $y = \underline{\hspace{2cm}}$

30-sec partner check: What is k for your Do Now rectangles (area = 144)?

DOK 1-2

33 min

student work

7 items in order. ~4–5 min per item. Compute xy for every table before answering.

Try It 1 · Two tables: inverse variation? Compute xy for each column. Is the product constant?

Practice #9 · Generalize From the table, how can you tell the data do NOT show inverse variation?

Practice #14 · Table check Do these values represent inverse variation? Explain.

Try It 2 · Write the equation $x = 6$ when $y = 1/2$. (a) Write the equation. (b)

DOK 2

5 min

academic conversation

Sentence frame · Pair share then whole class “As _____ increases, _____ decreases. Their product is _____, so the equation is $y = \text{_____}/x$.”

Class signals **thumbs up** (agree) / **sideways** (unsure).

Bridge to Period 2 Next class: **real-world inverse variation problems** (bouzouki string).

Also: DIRECT vs INVERSE — both can appear in the same context!

Period 2 carries the DOK-3 performance task.

DOK 1-2

2 min

not graded

Complete on your exit ticket

1. Today I learned that _____.
2. The difference between inverse and direct variation is _____.
3. One thing I'm still unsure about is _____.